Rahul Deepak Kuchipudi

Software Engineer – Edge AI Inference & Video Analytics

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EDUCATION

Indian Institute of Technology Bombay 2021 – 2025
B.Tech. in Computer Science and Engineering • JEE Advanced **AIR 709** • JEE Main **AIR 157** • **NTSE Scholar** Mumbai, India

EXPERIENCE

Software Development Engineer Jul 2025 – Present
Avathon (formerly SparkCognition) Bengaluru, India

IRIS Video Analytics Platform

- Core engineer on the IRIS rewrite from **DeepStream 6.2 to 9.0**; separated the C++/GStreamer core from cloud and use-case code through a stable plugin ABI, so modules ship independently. The runtime loads 3 shared libraries and 12 use-case modules.
- Increased per-GPU camera capacity **~2.7× at 25% activity** through per-camera activity gating. In a separate RTX PRO 6000 scale benchmark, sustained **99/99 active streams at 584 aggregate fps**, with a per-camera floor above 5 fps.
- Root-caused a **TensorRT 10.14 miscompile** of DEIMv2 deformable attention using an ONNX-vs-engine comparison; replaced GridSample ops with an MSDA plugin via ONNX graph surgery, restoring **98.7% person-count parity** with ONNX Runtime.
- Restored missing helmet alerts by tracing failed PPE-to-person association to a bounding-box geometry mismatch; corrected the association rule and added nine regression cases.
- Traced growing alert queues to a sender limited to one request per round trip; added 16 delivery workers with persistent libcurl connections. Delivery rose **15%** → **98%** at 0.5 s p50 lag in a 25-camera test with 250 ms simulated cloud latency.
- Diagnosed low GPU utilization by isolating stream-synchronization and batch-timeout settings; added a unit-tested 40 ms minimum timeout for synchronized inputs, restoring **94.9 aggregate fps across 25 cameras** with optical flow enabled.
- Fixed pipeline hangs when all sources stalled or never produced frames; added a unit-tested global-stall watchdog to catch failures missed by per-source checks.

VSS Vision-Language Systems & Edge Search

- Architected VSS, a vLLM-served vision-language layer (Cosmos-Reason2-2B) over Redis Streams/MQTT for scene descriptions and alert verification; built custom C++ GStreamer plugins with GPU-resident frame transfer. Scene description at **1.77 s p50 per 10 s window**.
- Raised VLM decode throughput **4.7**× (41 → 193 tok/s) and cut verdict latency **79%** (1.95 → 0.41 s) by enabling CUDA graphs in vLLM after profiling exposed launch overhead and 77% GPU idle time.
- Implementing edge video search with Cosmos-Embed1 scene embeddings, SigLIP2 object crops, LanceDB/SQLite and FastAPI; aligned crops to detection frame IDs to reject stale frames. Benchmarked scene search at **12 cameras @ 5 fps on one RTX PRO 4000**.

Software Developer Intern May 2024 – Jul 2024
Hilti Technology Solutions India Pune, India

- Automated MSI installer creation with GitLab CI/CD, Docker and WiX, cutting deployment from **2 hours to 10 minutes**; built Selenium/C# UI tests using the Page Object Model.

PROJECTS

Kirn — AI Terminal Python, Ollama, Local LLMs Open Source

- Built a Python terminal around bash and Ollama; used full-duplex PTYs to relay keyboard input and terminal output for interactive programs such as `vim` and `ssh`. Runs locally on Linux and Android/Termux.
- Implemented a tool-calling execution loop that returns command output to the model for follow-up actions and error recovery; stored successful tasks in persistent JSON memory for keyword-based lookup.

Church Live Translation Python, faster-whisper, IndicTrans2, WebSocket Jul – Aug 2026

- Built offline live captions with faster-whisper, IndicTrans2 and FastAPI/WebSockets; batched Malayalam, Telugu and Hindi translations in **one GPU call**, serving projector and phone displays within **~3–4 GB VRAM on an RTX 3060**.
- Fixed mistranslations from split speech by assembling sentences before translation and processing each as a separate batch row; logged transcripts and stage timings, replayed caption history after reconnects, and authenticated operator controls.

TECHNICAL SKILLS

Languages	Python, C++, C, Bash, SQL
Inference & Edge	NVIDIA DeepStream, TensorRT, CUDA, GStreamer, Triton, ONNX Runtime, FP16/INT8 Quantization
LLM & ML	vLLM, PyTorch, VLMs/LLMs, Hugging Face, OpenCV, NumPy, Pandas, Scikit-learn
Systems & Infra	Docker, Kubernetes, Kafka, Redis, MQTT, Linux, Git, GitLab CI/CD, meson/ninja
Certification	Deep Learning Specialization — DeepLearning.AI, Coursera (Nov 2025)